

ASSIGNMENT : 6

TITLE : Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement : Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives : To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory : An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling

Exercise :

1. Create a **Vehicle program** where an inner class displays vehicle details and anonymous class performs an action.

```
interface Action {
    void perform();
}

class Vehicle {
    String name = "Car";
    int speed = 120;

    class Details {
        void display() {
            System.out.println("Vehicle: " + name);
            System.out.println("Speed (km/h) :" + speed);
        }
    }
}

public class Main {
    public static void main(String[] args) {
        Vehicle v = new Vehicle();
        Vehicle.Details d = v.new Details();
        d.display();

        Action a = new Action() {
            public void perform() {
                System.out.println("Vehicle is moving.");
            }
        };

        a.perform();
    }
}
```

Output

Generated files

```
Vehicle: Car  
Speed (km/h) :120  
Vehicle is moving.  
|
```

2. Develop a **Food Delivery Application** where an inner class handles order details and anonymous classes handle delivery status updates.

```
interface DeliveryStatus {  
    void update();  
}  
  
class FoodOrder {  
  
    String orderId = "9401";  
    String item = "Pizza";  
  
    class OrderDetails {  
        void display() {  
            System.out.println("Order ID: " + orderId);  
            System.out.println("Item: " + item);  
        }  
    }  
}  
  
public class Base {  
    public static void main(String[] args) {  
  
        FoodOrder order = new FoodOrder();  
        FoodOrder.OrderDetails details = order.new OrderDetails();  
        details.display();  
  
        DeliveryStatus status = new DeliveryStatus() {  
            public void update() {  
                System.out.println("Delivery Status: Order Delivered");  
            }  
        };  
  
        status.update();  
    }  
}
```

```
Output    Generated files

Order ID: 9401
Item: Pizza
Delivery Status: Order Delivered
|
```

Outcome :

- Understood the concept and implementation of **inner classes**.
- Implemented **anonymous classes** for one-time object creation.
- Learned how these concepts improve encapsulation and simplify event handling.

Conclusion :

This assignment demonstrated the use of **inner classes** and **anonymous classes** in Java. Inner classes help logically group related classes and provide access to the members of the enclosing class, while anonymous classes simplify code by allowing one-time implementation of interfaces or abstract classes. These concepts are widely used in GUI programming, event handling, and object-oriented application development.